

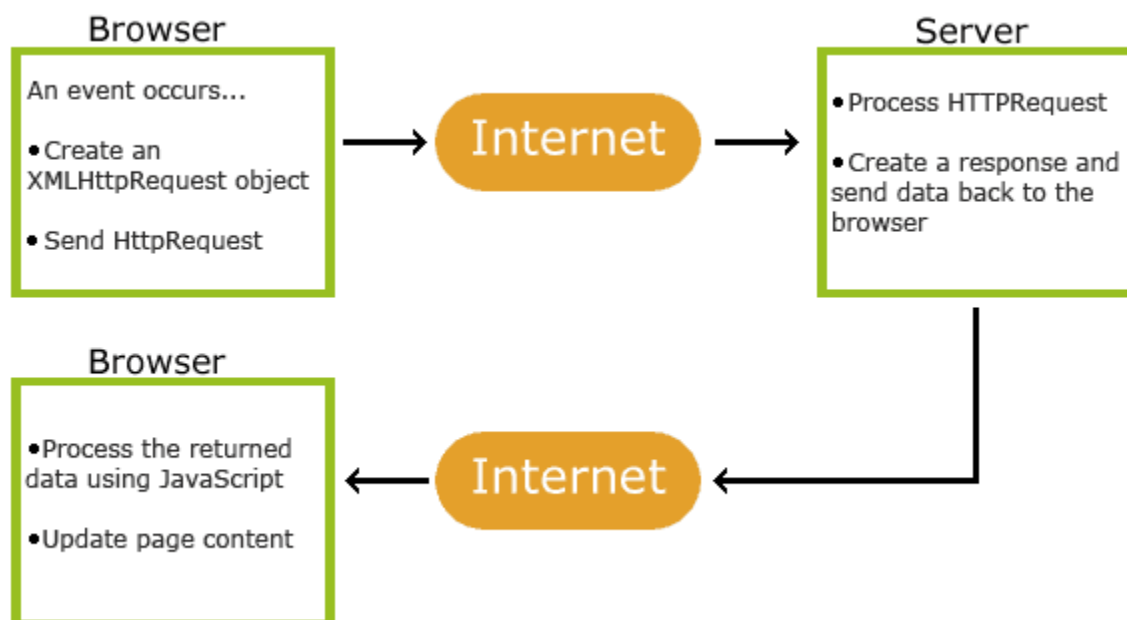
What is AJAX?

AJAX stands for asynchronous JavaScript and XML. AJAX is not a programming language or technology, but it is a combination of multiple web-related technologies like HTML, XHTML, CSS, JavaScript, DOM, XML, and the XMLHttpRequest object. The AJAX model allows web developers to create web applications that can interact with the user dynamically. It will also be able to quickly make a background call to web servers to retrieve the required application data. Then update the small portion of the web page without refreshing the whole web page.

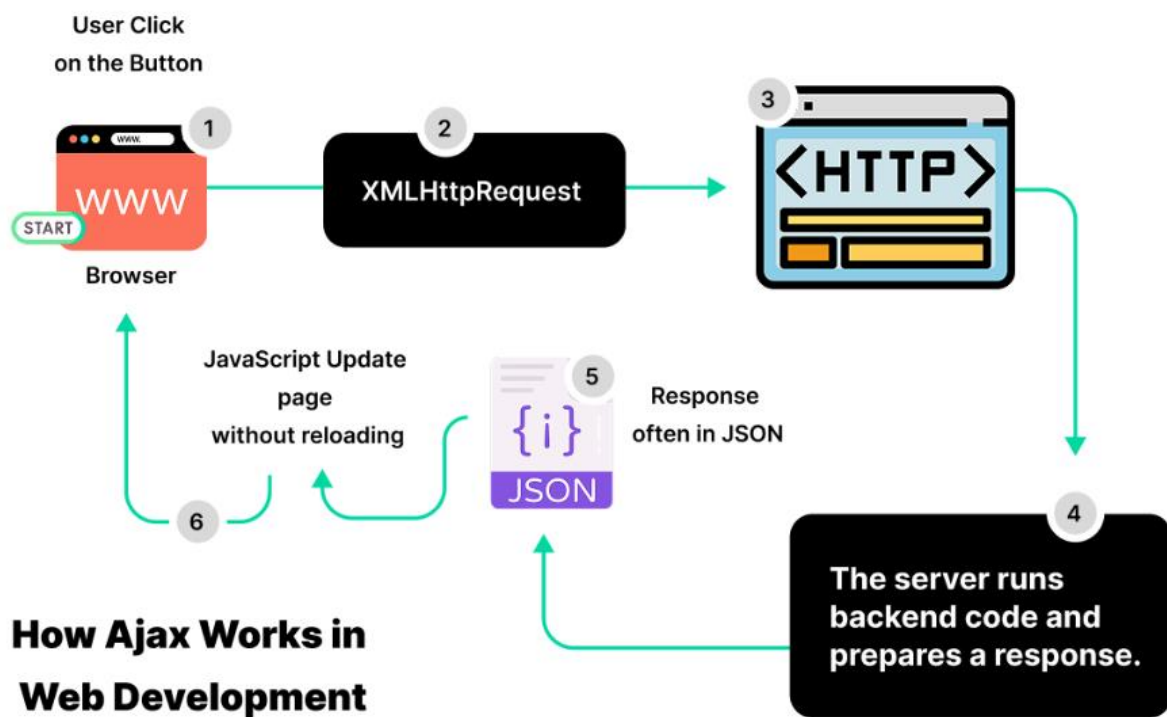
In the AJAX applications, the exchange of data between a web browser and the server is asynchronous means AJAX applications submit requests to the web server without pausing the execution of the application and can also process the requested data whenever it is returned. For example, Facebook uses the AJAX model, so whenever we like any post, the count of the like button increases instead of refreshing the whole page.

Working of AJAX

How AJAX Works



- i. An event occurs in a web page (the page is loaded, a button is clicked)
- ii. An XMLHttpRequest object is created by JavaScript
- iii. The XMLHttpRequest object sends a request to a web server
- iv. The server processes the request
- v. The server sends a response back to the web page
- vi. The response is read by JavaScript
- vii. Proper action (like page update) is performed by JavaScript



Example:

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>AJAX</title>
  <script src="script.js"></script>
```

```
</head>
<body>

<div id="demo">
  <h2>Open Console</h2>
</body>
</html>
```

Script.js

```
var xhttp = new XMLHttpRequest();
xhttp.open("GET", "info.json", true);
xhttp.onreadystatechange = function () {
  if (this.readyState == 4 && this.status == 200) {
    var response = JSON.parse(this.responseText);
    console.log(response.blog.name);
  }
};
xhttp.send();
```

info.json

```
{
  "blog": {
    "name": "My Blog",
    "posts": [
      {
        "title": "Python Basic",
        "url": "https://www.w3schools.com/python/"
      },
      {
        "title": "HTML Basics",
        "url": "https://www.w3schools.com/html/"
      }
    ]
  }
}
```

Details: <https://flatcoding.com/tutorials/javascript/what-is-ajax-how-it-works-in-web-development-with-examples/#section-use-xmlhttprequest-in-plain-javascript>

Another Example:

[Index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>AJAX Student Info</title>
  <script src="script.js"></script>
</head>
<body>

  <h2>Student Information</h2>
  <button onclick="loadStudents()">Load Students</button>

  <div id="output"></div>

</body>
</html>
```

[Script.js](#)

```
function loadStudents() {
  var xhr = new XMLHttpRequest();
  xhr.open("GET", "info.json", true);

  xhr.onreadystatechange = function () {
    if (xhr.readyState === 4 && xhr.status === 200) {
      var students = JSON.parse(xhr.responseText);

      var html = "<ul>";
      for (var i = 0; i < students.length; i++) {
        html += "<li>" + students[i].name + " - " + students[i].dept + "</li>";
      }
      html += "</ul>";

      document.getElementById("output").innerHTML = html;
    }
  };

  xhr.send();
}
```

info.json

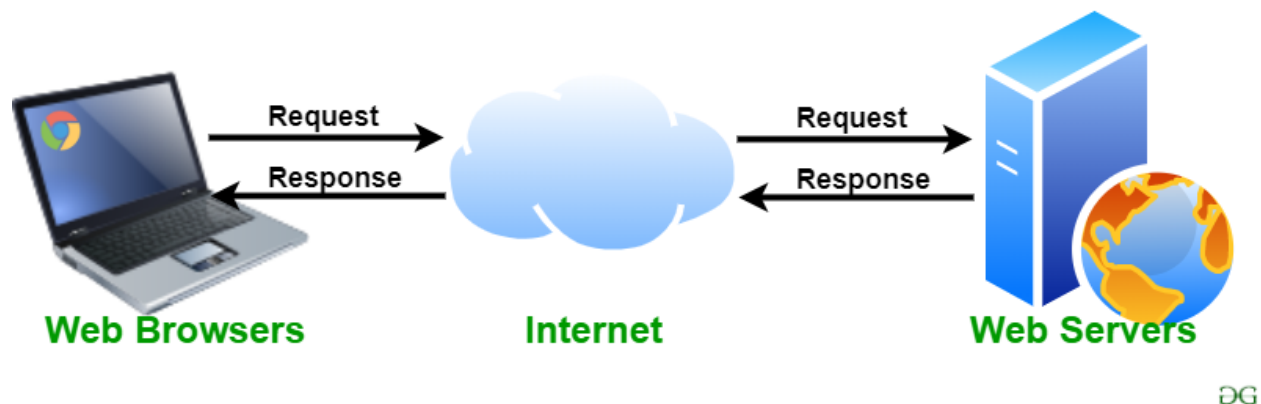
```
[  
  { "name": "Rahim", "dept": "ICT" },  
  { "name": "Karim", "dept": "CSE" },  
  { "name": "Fatima", "dept": "EEE" }  
]
```

Web Server:

CT A web server is a system—either software, hardware, or both—that stores, processes, and delivers web content to users over the Internet using the HTTP or HTTPS protocol. When a user's browser sends a request (like visiting a website), the web server responds by delivering the appropriate resources, such as HTML pages, images, videos, or data.

How Does a Web Server Work?

CT When a user accesses a website by entering a URL in their web browser, the browser sends an HTTP request to the web server hosting the website. The web server processes this request and returns the necessary resources to display the page on the user's browser.



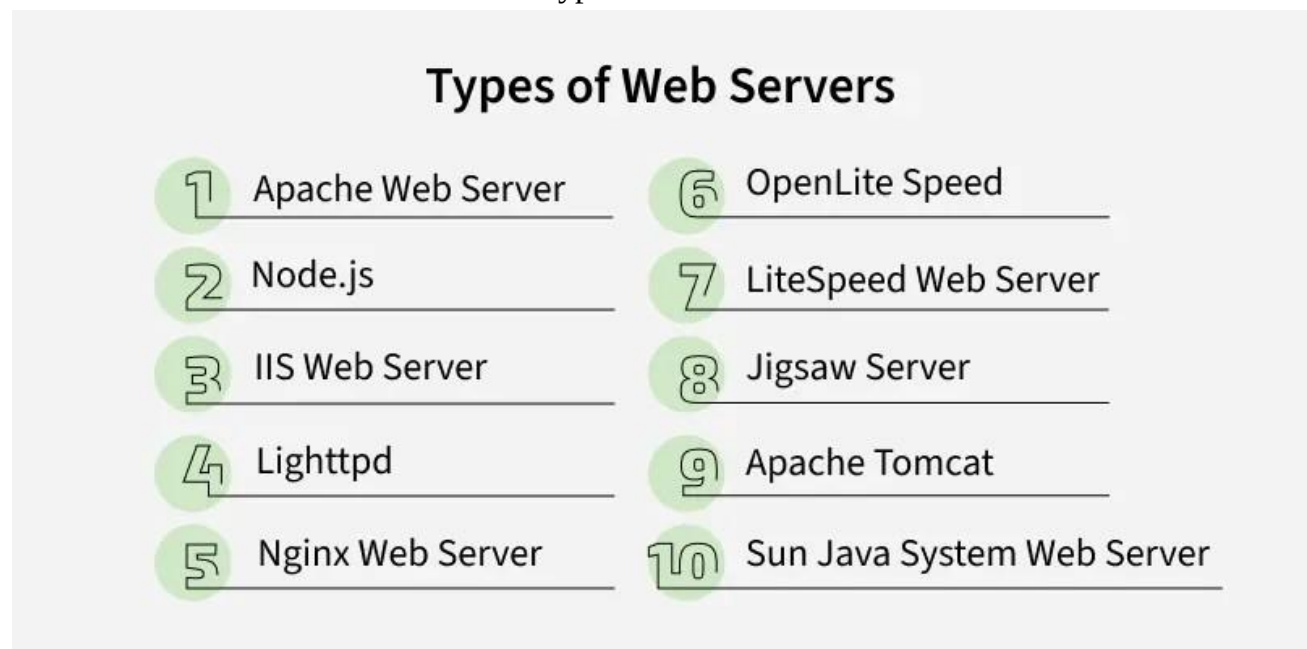
Here is a simplified version of how the process works

- i. **Client Request:** In the web browser (<https://www.example.com/>), the user enters a URL.
- ii. **DNS Resolution:** To get the IP address of the requested domain, the browser contacts a Domain Name System (DNS) server.
- iii. **Connecting to the Web Server:** Using the obtained IP address, the browser establishes a connection with the web server.

- iv. **Processing Request:** The web server receives the request and processes it.
- v. **Serving the Response:** The requested files (HTML, CSS, JavaScript, images) are sent back to the client's browser by the web server.
- vi. **Rendering the Web Page:** Based on the received data, the browser displays the web page to the user.

Types of Web Servers

Web servers can be categorized based on their functionality, usage, and implementation. Below are some of the most common types:



Which Web Server Should You Use and When?

Choosing the right web server depends on what you need for your website or application. Here's a simple guide to help you decide:

1. **Use Apache:** If you want a reliable and customizable web server that works on almost any system. It's great for general websites and supports many features. Apache is a fundamental part of LAMP (Linux, Apache, MySQL, PHP), the most popular web development stack. It supports easy integration with database management systems such as MySQL and scripting languages such as PHP and Python to develop dynamic web applications.

LAMP stands for:

Linux: An open-source operating system that manages system resources and executes applications efficiently.

Apache: A widely used web server that processes HTTP requests and serves web content securely.

MySQL: A relational database management system (RDBMS) that stores and retrieves structured data efficiently.

PHP: A server-side scripting language that integrates with Apache to generate dynamic web content.

Apache is the foundation of most **CMS (Content Management Systems)** such as WordPress, Joomla, and Drupal, and thus is a favorite among hosting dynamic websites.

2. **Use Nginx:** If your website gets a lot of visitors and you need a fast and efficient server that can handle high traffic smoothly.
3. **Use IIS:** If you are using Windows-based applications and need a server that works well with Microsoft technologies like ASP.NET.
4. **Use LiteSpeed:** If you want a faster and more secure alternative to Apache, especially for WordPress or other PHP-based websites.
5. **Use Apache Tomcat:** If your website or app is built with Java and you need a server that supports Java Servlets and JSP (Java Server Pages).
6. **Use NodeJS:** If you're building real-time applications, such as chat apps or online games, and want to use JavaScript for both frontend and backend.
7. **Use Lighttpd:** If you need a lightweight and fast server that works well on systems with low memory or limited resources.
8. **Use OpenLiteSpeed:** If you want a free version of LiteSpeed that still offers great speed and performance, etc.

Local Development Servers

XAMPP = Apache + MySQL + PHP + Perl in one package

WAMP = Windows-based Apache + MySQL + PHP

MAMP = Mac-based Apache + MySQL + PHP

Laragon = Lightweight, portable dev environment for Windows

Other example: not for exam

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>AJAX Dropdown Example</title>

  <script src="script.js"></script>

</head>

<body>

  <h2>Select Department</h2>

  <select onchange="loadStudents(this.value)">

    <option value="">-- Select Department --</option>

    <option value="ICT">ICT</option>

    <option value="CSE">CSE</option>

    <option value="EEE">EEE</option>

  </select>

  <div id="output"></div>

</body>

</html>

-----

function loadStudents(dept) {

  if (dept === "") {

    document.getElementById("output").innerHTML = "";

    return;

  }

}
```

```

}

var xhr = new XMLHttpRequest();

xhr.open("GET", "info.json", true);

xhr.onreadystatechange = function () {

    if (xhr.readyState === 4 && xhr.status === 200) {

        var students = JSON.parse(xhr.responseText);

        var html = "<ul>";

        for (var i = 0; i < students.length; i++) {

            if (students[i].dept === dept) {

                html += "<li>" + students[i].name + "</li>";

            }

        }

        html += "</ul>";

        document.getElementById("output").innerHTML = html;

    }

};

xhr.send();

}

```

```

-----

[

    { "name": "Rahim", "dept": "ICT" },

    { "name": "Karim", "dept": "CSE" },

    { "name": "Fatima", "dept": "EEE" },

    { "name": "Hasan", "dept": "ICT" }

]

```